

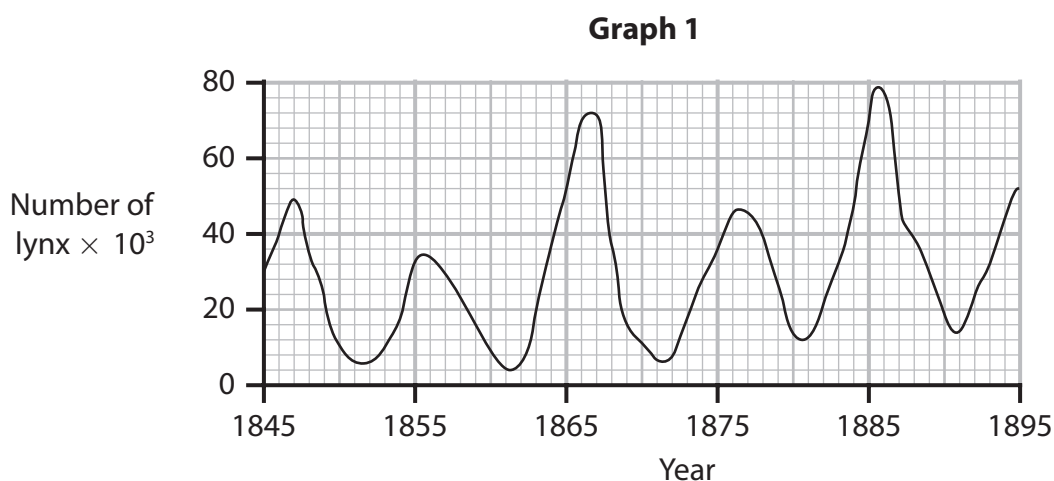
3 The photograph shows a Canadian lynx and a hare.



(Source: © FLPA / Alamy Stock Photo)

The lynx is a type of cat and is a predator of the hare.

Graph 1 shows the changes in the number of lynx from 1845 to 1895.



(a) (i) In 1863, there were 150 000 hares.

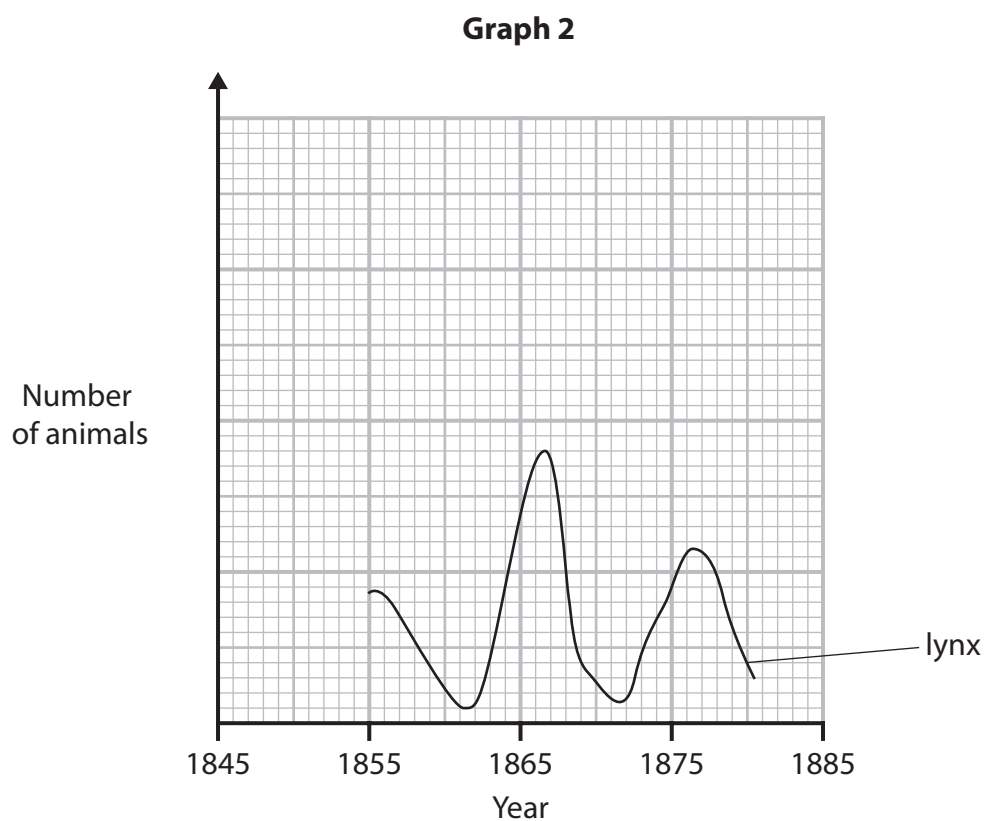
Calculate how many times more hares there were than lynx.

(1)

Answer

(ii) Graph 2 shows part of graph 1.

Sketch a line on graph 2 to show the likely changes in the numbers of hares from 1855 to 1880.



(2)

(iii) Explain the changes in the numbers of lynx and hares.

(3)

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(b) In many ecosystems, the relationship between one predator species and one prey species is less predictable.

Suggest **two** reasons why this relationship is not always predictable.

(2)

(Total for Question 3 = 8 marks)
